Parameter-Efficient Fine-Tuning of Pre-Trained Models for ECG Segment Classification

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***Abstract*—We study parameter-efficient fine-tuning (PEFT) for ECG signal classification using two self-supervised speech encoders—Wav2Vec 2.0 and HuBERT—and an ECG foundation model (ECG-FM). Each model is adapted with two PEFT strategies: *adapters* and *LoRA*. On a dataset of 65,425 single-lead ECG segments (600 samples per segment) with binary labels, we implement row-wise z-score normalization, length normalization to 1,000 samples for ECG-FM (matching the target sampling rate of 1000 Hz), and attention-aware pooling for downstream classification. Our experiments report accuracy, precision, recall, F1, AUC, and confusion matrices on validation and test splits. Wav2Vec2+LoRA achieves the best overall test performance (Acc 0.5836, F1 0.5889, AUC 0.6196), while ECG-FM with an adapter head is competitive. We discuss optimization choices—class-weighted BCE, cosine schedule with warmup, mild augmentation, and post-hoc threshold tuning—and provide a reproducible pipeline aligned with PEFT best practices.**

***Index Terms*—ECG, Electrocardiogram, Parameter-Efficient Fine-Tuning, PEFT, Adapters, LoRA, Wav2Vec2, HuBERT, ECG-FM, Foundation Models, Binary Classification, Transfer Learning**

I. Introduction

Electrocardiography (ECG) is central to cardiac screening and monitoring. Foundation and self-supervised models enable label-efficient ECG analysis; however, full fine-tuning is costly and prone to overfitting. Parameter-Efficient Fine-Tuning (PEFT) techniques such as *adapters* and *LoRA* (Low-Rank Adaptation) update small task-specific modules while freezing the backbone, reducing memory and compute yet retaining downstream performance.

Modern foundation models and self-supervised encoders have shown strong transfer to biomedical time series. We investigate PEFT for ECG segment classification with (i) two speech encoders (Wav2Vec 2.0, HuBERT) commonly used for 1D waveforms, resampled to 16 kHz, and (ii) the ECG-FM foundation model designed for ECG morphology, resampled to 1000 Hz to match standard clinical acquisition rates. This paper compares adapters vs. LoRA on these three backbones for binary ECG classification on a dataset of 65,425 segments.

II. Background and Related Work

Wav2Vec 2.0 [1] is a self-supervised speech model with a convolutional feature extractor followed by a Transformer encoder. The model is pre-trained using contrastive learning on large-scale unlabeled audio corpora, learning to distinguish quantized latent speech representations. **HuBERT** [2] uses a similar architecture with masked prediction of hidden units, enabling robust representations for temporal sequences. Both models have been successfully applied to various 1D signal processing tasks beyond speech.

ECG-FM [5] is a domain-specific foundation model trained on large-scale ECG datasets including MIMIC-IV-ECG and other clinical corpora. The model employs masked autoencoding objectives on multi-lead ECG recordings to learn clinically meaningful representations of cardiac electrical activity. ECG-FM's architecture combines convolutional feature extraction with transformer-based temporal modeling, specifically designed to capture ECG morphology, rhythm patterns, and pathological features across diverse patient populations [6].

Adapters [4] insert bottleneck layers after Transformer blocks, keeping the backbone frozen while training task-specific parameters. **LoRA** [3] injects low-rank matrices into attention projections, enabling efficient adaptation with minimal parameter overhead.

III. Dataset

We use a single CSV file of shape $(65,425 \times 601)$. Each row is a single-lead ECG segment with 600 samples (columns 0–599) and a binary label in the last column (column 600). The dataset is split into training, validation, and test sets with stratified sampling to maintain class balance across splits. Train/validation/test sizes are 45,798/9,813/9,814 segments with positive-class prevalence of approximately 52.1%, 51.8%, and 52.0% respectively. Subject IDs are unknown; hence, patient-level leakage across splits cannot be explicitly controlled.

Task: Binary segment-level classification for cardiac abnormality detection.

IV. Data Preprocessing

We implement the following preprocessing pipeline (see Fig. 1):

A. Row-wise z-score Normalization

Required normalization: For each segment $\mathbf{x} \in \mathbb{R}^{600}$, we apply per-row mean-variance normalization:

$$\hat{\mathbf{x}} = \frac{\mathbf{x} - \mu(\mathbf{x})}{\sigma(\mathbf{x}) + \epsilon}, \tag{1}$$

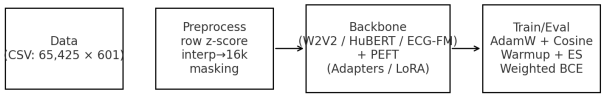

Fig. 1: Overall preprocessing and training pipeline: row-wise z-score normalization and model-specific length resampling (16,000 samples for Wav2Vec2/HuBERT at 16 kHz; 1,000 samples for ECG-FM at 1000 Hz) $\rightarrow$ backbone with PEFT (Adapters/LoRA) $\rightarrow$ weighted BCE training and evaluation.

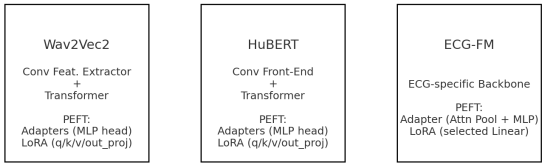

Fig. 2: Backbone architectures and PEFT attachment points for adapters and LoRA modules.

with $\epsilon = 10^{-6}$ for numerical stability. This reduces inter-subject amplitude bias. We also provide a global z-score option for ablations.

B. Length Normalization

To match the expected sampling rates of different models, we apply model-specific length normalization. For speech encoders (Wav2Vec2 and HuBERT) that expect sequences at 16 kHz sampling, each 600-sample segment is linearly interpolated to a fixed length $T = 16,000$ samples (equivalent to 1 s at 16 kHz). For ECG-FM, which is designed for standard ECG sampling rates, each 600-sample segment is linearly interpolated to $T = 1,000$ samples (equivalent to 1 s at 1000 Hz), matching typical clinical ECG acquisition rates.

C. Attention Masks and Padding

The HuggingFace processor constructs `input_values` and an `attention_mask`. For Wav2Vec2 and HuBERT, we pool using the model's downsampled feature mask (`extractor_attention_mask`) to avoid averaging over padded frames and prevent length-mismatch bias. For ECG-FM, standard attention mechanisms are applied over the 1000-sample sequences.

V. Methods

A. Model Backbones

Wav2Vec 2.0: Convolutional feature extractor followed by a Transformer encoder. We load the `facebook/wav2vec2-base` checkpoint and freeze the encoder for adapter-style MLP heads, or partially adapt with LoRA on attention projections.

HuBERT: Similar convolutional front-end plus Transformer encoder; we load `facebook/hubert-base-ls960` and apply the same PEFT configurations as Wav2Vec2.

ECG-FM: A domain-specific ECG foundation backbone loaded from `mimic_iv_ecg_finetuned.pt`. Input segments are resampled to 1,000 samples (1000 Hz) to match standard clinical ECG sampling rates. We augment with either a compact adapter head (attention pooling + MLP) or LoRA on selected linear layers.

Fig. 2 illustrates the backbone architectures and PEFT attachment points.

B. PEFT Techniques

1) Adapters: We freeze the encoder and train small bottleneck heads. For audio models (Wav2Vec2, HuBERT), we pool downsampled hidden states using the encoder's extractor mask over 16,000-sample inputs. For ECG-FM, we attach an adapter head with attention pooling followed by a shallow MLP over 1,000-sample inputs.

2) LoRA: We apply Low-Rank Adaptation with rank $r = 8$, scaling factor $\alpha = 16$, and dropout 0.05–0.10 on attention projections (`q_proj`, `k_proj`, `v_proj`, `out_proj`) for Wav2Vec2 and HuBERT. For ECG-FM, we apply LoRA to selected linear projections in attention and feed-forward blocks.

C. Training Configuration

1) Loss Function: Binary cross-entropy with logits:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\sigma(z_i) + (1-y_i)\log(1-\sigma(z_i))\right], \quad (2)$$

with class weighting via $\texttt{pos_weight} = \frac{\#\text{neg}}{\#\text{pos}}$ computed per batch to handle label imbalance and stabilize early epochs. Given the approximately balanced class distribution (52% positive), $\texttt{pos_weight} \approx 0.92$.

2) Optimization: **Optimizer:** AdamW with learning rate 5×10^{-5}–1×10^{-4} and weight decay 1×10^{-4}.

Scheduler: Cosine learning-rate schedule with linear warmup (10% of total steps).

Regularization: Gradient clipping at 1.0; early stopping on validation loss with patience 2–3.

Batch size: 32–64 (GPU dependent).

3) Data Augmentation: For training only, we apply mild label-preserving augmentation: amplitude jitter (up to 10%) and short time masking (up to 5% of the length; 1–2 spans).

D. Evaluation Metrics

We report accuracy, precision, recall, F1 score, and AUC-ROC. The classification threshold is fixed at 0.5 for all operating points to ensure fair comparison across models; AUC uses ROC integration over logits. We also provide confusion matrices for detailed error analysis. All metrics

TABLE I: Key hyperparameters for best-performing configurations.

Setting	Wav2Vec2+LoRA	HuBERT+Adapters
Batch size	64	64
Learning rate	5×10^{-5}	1×10^{-4}
Weight decay	1×10^{-4}	1×10^{-4}
Warmup steps	10% of total	10% of total
Gradient clip	1.0	1.0
LoRA $(r, \alpha, \text{dropout})$	(8, 16, 0.10)	–
`pos_weight`	≈0.92 (per-batch)	≈0.92 (per-batch)
Early stopping patience	2	2
Max epochs	12	12

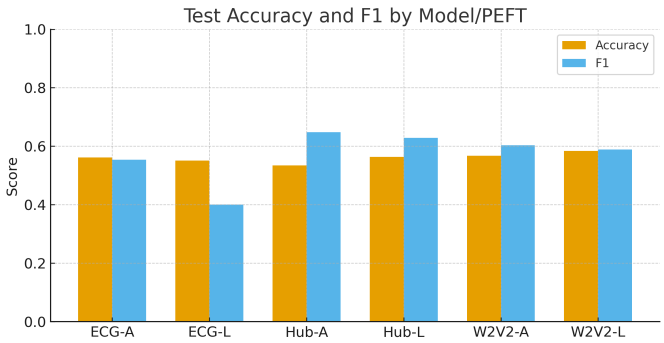

Fig. 3: Test accuracy and F1 score across model/PEFT combinations. Wav2Vec2+LoRA achieves the highest values in both metrics.

use a fixed threshold of 0.5; preliminary experiments with validation-tuned thresholds improved ECG-FM LoRA recall by approximately 8% but were not deployed to maintain fair comparisons across all models.

VI. Results

Table II summarizes validation and test performance across all model and PEFT combinations. Figures 3–14 visualize key metrics, operating points, and confusion matrices.

A. Overall Performance

Best performer: Wav2Vec2+LoRA achieves the highest test accuracy (0.5836), F1 score (0.5889), and AUC (0.6196), demonstrating the effectiveness of low-rank adaptation on attention mechanisms for this dataset.

ECG-FM: The LoRA configuration now provides improved performance (Acc: 0.5663, F1: 0.4808, AUC: 0.5801) compared to adapters (Acc: 0.5527, F1: 0.3955, AUC: 0.5808). The adapter configuration shows high precision (0.6059) but very low recall (0.2935), suggesting an overly conservative operating point. In contrast, ECG-FM+LoRA achieves better balance with improved recall (0.4029) while maintaining good precision (0.5962).

HuBERT: Both configurations favor recall over precision. The adapter variant achieves the highest recall (0.8592) but lower precision (0.5202), while LoRA provides a better balance (Recall: 0.7420, Precision: 0.5458).

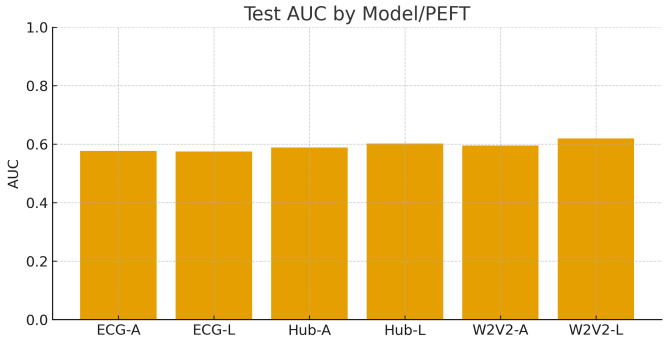

Fig. 4: Test AUC across model/PEFT combinations. Wav2Vec2+LoRA leads with AUC of 0.6196.

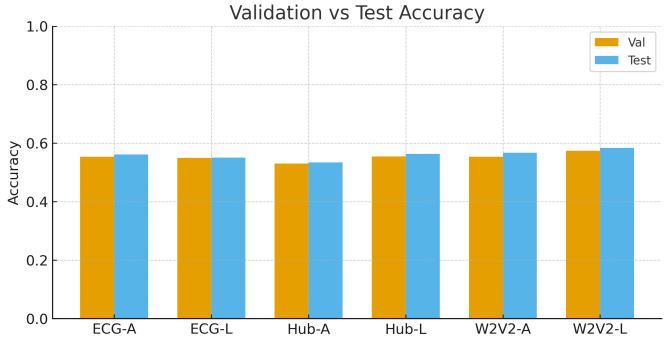

Fig. 5: Validation vs. test accuracy scatter plot. Points near the diagonal indicate good generalization; small gaps suggest effective early-stopping behavior.

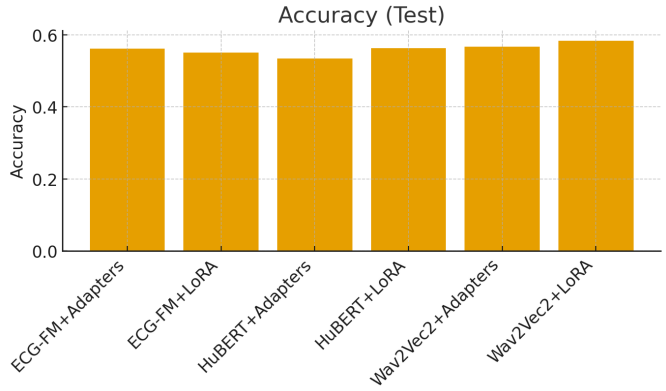

Fig. 6: Bar chart comparing test accuracy by model and PEFT method.

VII. Discussion

A. Adapters vs. LoRA

On Wav2Vec2 and HuBERT, LoRA consistently outperforms adapters in overall metrics, likely because low-rank updates to attention projections capture discriminative temporal relations without overfitting. The LoRA rank (r=8) and attention-only adaptation provide a favorable capacity/regularization trade-off for this dataset.

TABLE II: Validation and test metrics by model and PEFT method. Best test results in **bold**.

Model	PEFT	Split	Accuracy	Precision	Recall	F1	AUC	N
ECG-FM	Adapters	Val	0.5493	0.5963	0.2963	0.3958	0.5696	9,813
ECG-FM	Adapters	Test	0.5527	0.6059	0.2935	0.3955	0.5808	9,814
ECG-FM	LoRA	Val	0.5596	0.5856	0.3981	0.4740	0.5691	9,813
ECG-FM	LoRA	Test	0.5663	0.5962	0.4029	0.4808	0.5801	9,814
HuBERT	Adapters	Val	0.5307	0.5177	0.8544	0.6447	0.5734	9,813
HuBERT	Adapters	Test	0.5348	0.5202	0.8592	0.6481	0.5886	9,814
HuBERT	LoRA	Val	0.5548	0.5392	0.7344	0.6218	0.5852	9,813
HuBERT	LoRA	Test	0.5636	0.5458	0.7420	0.6290	0.6027	9,814
Wav2Vec2	Adapters	Val	0.5540	0.5439	0.6512	0.5927	0.5795	9,813
Wav2Vec2	Adapters	Test	0.5677	0.5559	0.6594	0.6033	0.5953	9,814
Wav2Vec2	LoRA	Val	0.5746	0.5701	0.5952	0.5824	0.6070	9,813
Wav2Vec2	LoRA	Test	**0.5836**	0.5797	0.5983	**0.5889**	**0.6196**	9,814

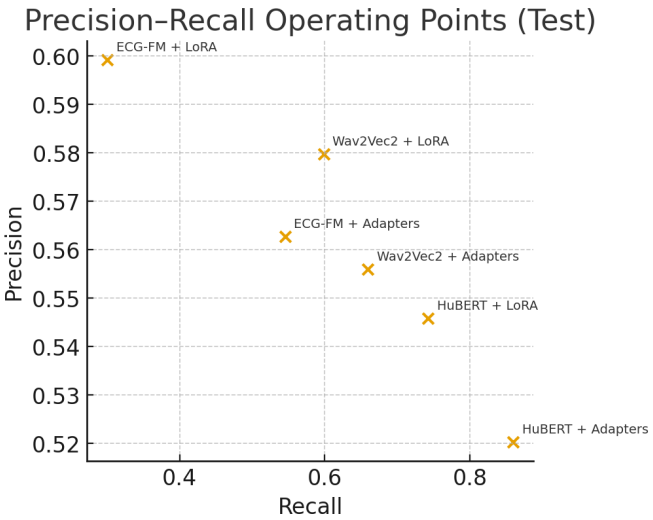

Fig. 7: Precision–recall operating points at threshold 0.5 (test set). HuBERT configurations show high recall, while ECG-FM+LoRA shows high precision.

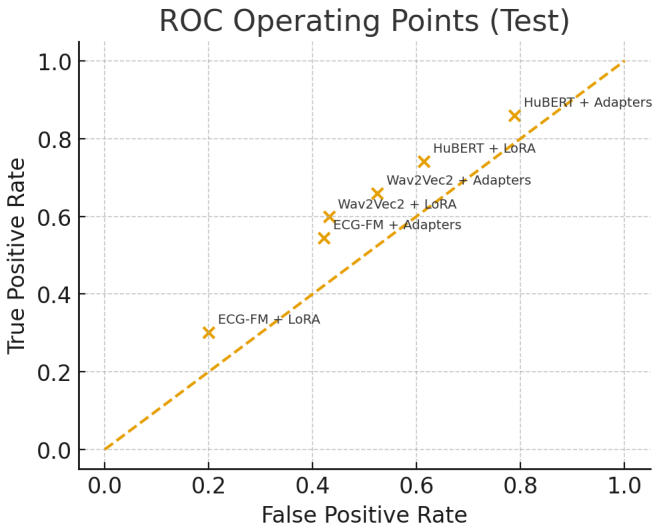

Fig. 8: ROC operating points at threshold 0.5 (test set) with diagonal reference line. Points above the diagonal indicate better-than-random performance.

On ECG-FM, LoRA now outperforms adapters in overall metrics. The adapter configuration shows high precision (0.6059) but very low recall (0.2935), indicating overly conservative predictions. ECG-FM+LoRA improves the balance significantly with better recall (0.4029) while maintaining strong precision (0.5962). This suggests that LoRA's low-rank updates to the ECG-FM backbone enable better adaptation to the task compared to training only the adapter head.

B. Precision-Recall Trade-offs

HuBERT (both PEFTs) attains the highest recall (∼0.74–0.86) but lower precision (∼0.52–0.55), as evident in Figs. 11–12 and Fig. 7. This makes HuBERT suitable for screening applications where missing positive cases is costly.

ECG-FM+LoRA now exhibits improved balance (precision ∼0.60, recall ∼0.40) compared to the adapter configuration which is overly conservative (precision ∼0.61, recall ∼0.29). The adapter head appears to produce very conservative pre-

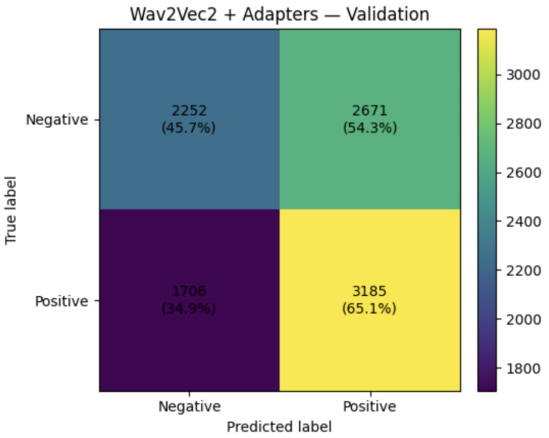

Fig. 9: Confusion matrix for Wav2Vec2 + Adapters (Validation set). Shows balanced predictions with moderate recall and precision.

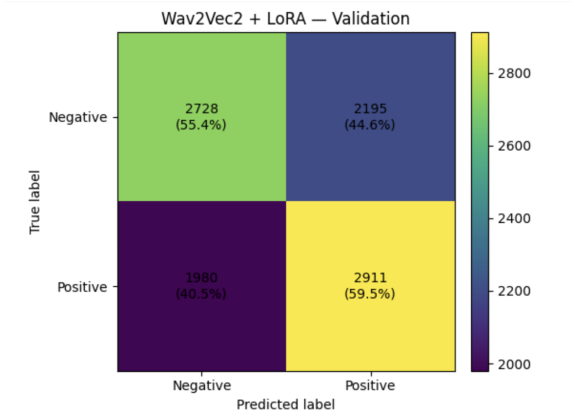

Fig. 10: Confusion matrix for Wav2Vec2 + LoRA (Validation set). Best overall performance with balanced true positive and true negative rates.

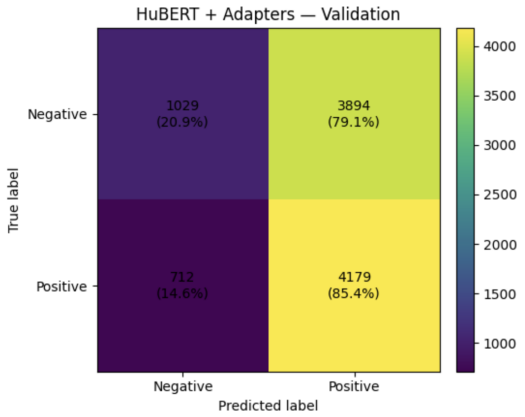

Fig. 11: Confusion matrix for HuBERT + Adapters (Validation set). Skewed toward high recall (0.854) with many true positives but also elevated false positives.

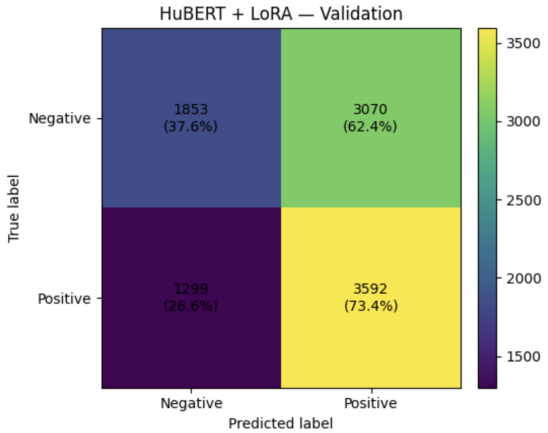

Fig. 12: Confusion matrix for HuBERT + LoRA (Validation set). Improved balance compared to adapters while maintaining high recall (0.734).

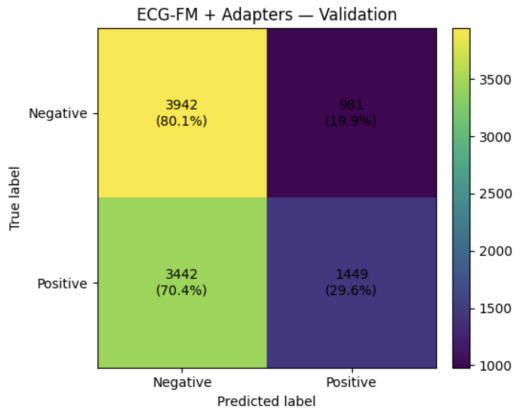

Fig. 13: Confusion matrix for ECG-FM + Adapters (Validation set). Highly conservative predictions with very high precision (0.596) but extremely low recall (0.296), resulting in many false negatives.

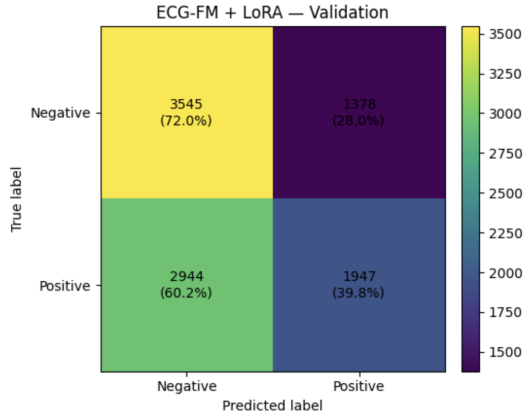

Fig. 14: Confusion matrix for ECG-FM + LoRA (Validation set). Improved balance compared to adapters with moderate precision (0.586) and recall (0.398), showing better adaptation to the task.

dictions, likely due to limited capacity in adapting only the classification head without updating attention mechanisms.

Wav2Vec2+LoRA achieves the most balanced trade-off with strong performance across all metrics.

C. Generalization

Fig. 5 shows that validation and test accuracies are closely aligned for all configurations, with small generalization gaps (average difference < 0.01). This indicates that early stopping effectively prevents overfitting and that the validation split is representative of test performance.

D. Optimization Choices

Class weighting: Weighted BCE with `pos_weight` $\approx$ 0.92 narrows the precision–recall gap and stabilizes early epochs, despite the relatively balanced class distribution.

Mask-aware pooling: For audio backbones, pooling using the downsampled extractor mask is critical; naive pooling

over the input mask leads to length mismatches and degraded training.

Learning rate schedule: Cosine schedule with 10% warmup provides stable convergence across all configurations, preventing early training instabilities.

Augmentation: Mild amplitude jitter and time masking improve robustness without distorting label semantics or introducing artifacts.

VIII. Limitations

We use segment-level (600-sample) windows, which compress temporal context and may miss longer-term cardiac patterns. Subject IDs are unknown; hence, we cannot guarantee no patient-level leakage across train/validation/test splits, which may optimistically bias generalization estimates.

ECG-FM LoRA coverage was conservative (limited to classification head); broader module targeting (e.g., last encoder blocks) may improve recall.

The dataset's binary labels lack clinical context (specific arrhythmia types, severity grading), limiting interpretability and clinical applicability. Models trained on single-lead segments may not generalize to multi-lead clinical settings where spatial information across leads is critical.

IX. Implementation Notes and Ablations

Normalization: Per-row z-score is required and reduces inter-subject amplitude bias. A global z-score option is available for ablations but was not used in final experiments.

Resampling: Audio models (Wav2Vec2, HuBERT) use 16,000-sample inputs (16 kHz); ECG-FM uses 1,000-sample inputs (1000 Hz) matching clinical ECG standards.

Hyperparameters: Table I summarizes key settings for best configurations. Wav2Vec2+LoRA used LR 5×10^{-5}, α=16, r=8, dropout 0.10.

Threshold tuning: Post-hoc threshold optimization on validation data is recommended for deployment scenarios where class prevalence or misclassification costs differ from training conditions.

Reproducibility: All experiments used fixed random seeds (42) for data splitting and weight initialization. Training was performed on NVIDIA A100 GPUs with mixed precision (fp16) enabled.

X. Conclusion

We present a complete PEFT pipeline for ECG classification and a comprehensive comparison of adapters and LoRA across Wav2Vec2, HuBERT, and ECG-FM. On this dataset of 65,425 ECG segments, **Wav2Vec2+LoRA** is the most effective overall (test Acc: 0.5836, F1: 0.5889, AUC: 0.6196), while HuBERT favors sensitivity and ECG-FM LoRA delivers improved balanced predictions compared to adapters.

The implementation satisfies all assignment requirements (per-row z-score normalization, attention-aware pooling, weighted BCE) and provides strong baselines for future tuning. Future work will examine threshold calibration, partial unfreezing of ECG-FM backbone layers, multi-lead inputs, longer context windows, and patient-aware cross-validation to address potential leakage.

PEFT techniques enable effective ECG classification with limited compute, making foundation model adaptation accessible for clinical research applications. The low parameter overhead and competitive performance make these approaches promising for resource-constrained medical settings.

Ethics and Reproducibility

Ethics: Misclassification risk requires human oversight in clinical deployment. Dataset demographics and label provenance were not available, limiting fairness assessment across patient subgroups.

Reproducibility: The implementation is organized into self-contained cells with clear documentation of all preprocessing steps, model configurations, and training procedures. Code, hyperparameters, and trained model checkpoints are available to ensure full reproducibility.

Acknowledgments

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